Sound: pitch and amplitude

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a drawing or explanation of a longitudinal wave to distinguish pitch from sound level.

SUCCESS CRITERIA - I can...

1. Describe compressions and rarefactions.
2. Connect pitch primarily with frequency.
3. Separate particle motion from sound propagation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “pitch” and “amplitude” to explain your answer. Sentence starters are optional.

pitch: the perceived highness or lowness of a sound

amplitude: maximum displacement from equilibrium for the wave

longitudinal: a disturbance parallel to the wave’s travel direction

01 Warm-up: bring back an earlier idea

Light moves from a transparent medium with n=**1.6** into air (n=**1**).
1. **Find critical angle** using sin θ_c=**1**/n.
2. **Describe incidence above it** in this ideal case.

Word help: medium = the material through which a mechanical wave travels

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: T = **1**/f; v = f λ.
Higher f generally means higher pitch. Amplitude affects intensity with the other parts of the setup kept the same, not frequency.

Worked reasoning: A tuning fork makes **440 cycles** each second, so its frequency is **440 Hz**. A louder version can have the same frequency but greater amplitude.

Your turn: Draw two displacement-time traces with equal periods and different amplitudes.

Sentence starter: Both ... . They differ in ... because ... .

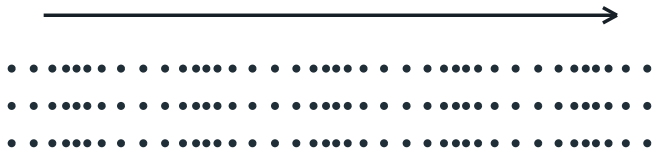


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Mark compressions and rarefactions in the particle picture. Do air particles travel from speaker to listener with the disturbance?

Sentence starter: I notice ... . This supports ... because ... .



Label dense and sparse regions.

Marzano check - circle: ① ② ③

04 Apply the physics

A tone generator plays **250 Hz** and **500 Hz** in the same air.
1. Use a sound speed of **340 m/s** to **calculate both wavelengths**.
2. Separately, double the pressure amplitude while frequency stays fixed. Use $I \propto A^2$ to **compare intensity**.
3. **Label which feature changes** in each diagram.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Turning up a speaker’s volume does not normally double a song’s pitch. **Explain which wave property mainly** changes.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Bodies initially at **340 K** and **360 K** exchange heat in an isolated system. **Explain what happens** to total energy and to the temperature-difference opportunity for work at equilibrium.
Label energy entering, leaving and stored.

Word help: heat = energy transferred because of a temperature difference

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.

Chosen system

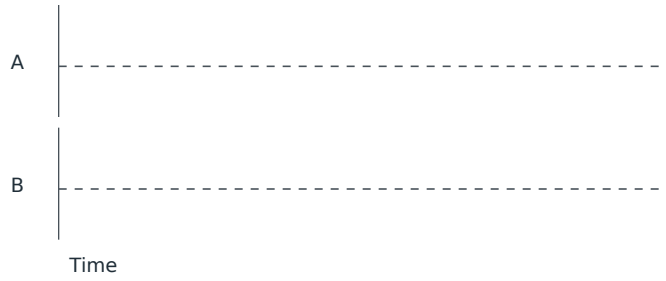
Stored energy change: _____ J

Marzano check - circle: ① ② ③

07 Exit check

A **500 Hz** tone lasts **0.10 s**. **Find number of cycles** and period.

Sentence starter: I found … [unit]. This value means … in this situation.

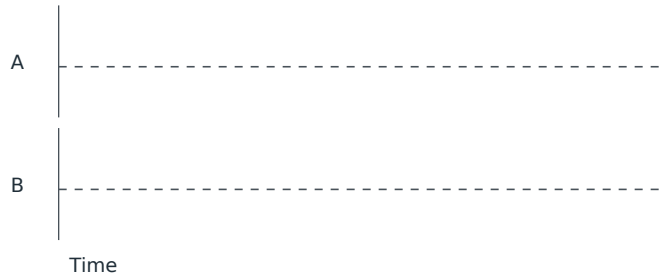


Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Two tones have frequencies **400** and **405 Hz**. Using beat frequency as their difference, find the beat rate. **Explain why beats** are not a new **405**-minus **400 Hz** pitch from either source alone.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③